

Qianqian Tong

PhD Candidate & Graduate Research Assistant
Maseeh Department of Civil, Architectural, and Environmental Engineering
The University of Texas at Austin
Ernest Cockrell Jr. Hall (ECJ) 6.3
301 E. Dean Keeton St.
Austin, TX, 78712-1172

Email: qianqian_tong@utexas.edu

Website: qianntong.github.io

Summary

PhD candidate in Transportation Engineering with experience in operations research, geospatial modeling, data analytics, route and network optimization, simulation, and machine learning. Experienced in developing data-driven decision-support solutions for rail, freight, multimodal transportation, and logistics systems, with industry experience in transportation, technology and healthcare. Hands-on experience with GIS, AI/ML methods, and open-source Python frameworks. Seeking a Spring 2027 internship or a full-time position starting after May 2027.

Education

University of Texas at Austin

Ph.D. in Transportation Engineering

Austin, TX

August 2023 – May 2027

Shanghai University of Engineering Science

B.E. in Transportation Engineering, Graduated with honors

Shanghai, China

September 2019 – July 2023

Research Experience

University of Texas at Austin

PhD Candidate, Intermodal Transportation Systems

August 2023 – Present

Austin, TX

- Developed an optimization–simulation framework for large-scale rail–road freight systems, integrating MILP-based terminal design, congestion simulation, and energy-aware resource allocation to evaluate terminal configurations and minimize infrastructure footprint under operational constraints.
- Developed queuing-based node and GIS/DEM-based link performance models to extract terminal layout/corridor spatial features, estimate capacity, congestion, processing time, travel time, and energy consumption across large-scale intermodal freight networks; manuscript accepted in *Journal of Rail and Rapid Transit*.
- Developed an intermodal freight competitiveness and pricing framework to evaluate capacity availability, optimal batch sizes for *Parallel Systems* autonomous rail vehicles, truck–rail break-even conditions, profit-maximizing rail pricing, and vehicle acquisition price thresholds under cost uncertainty; manuscript submitted to *Transportation Research Part A: Policy and Practice*.
- Implemented large-scale freight assignment and routing models, including shortest-path and user-equilibrium formulations, to evaluate container routing and system performance under varying demand, capacity, infrastructure, and disruption scenarios.

Texas Department of Transportation

Research Assistant, Multimodal Analysis and Routing Model for Texas

January 2026 – December 2026

Austin, TX

- Developed data pipelines and an interactive dashboard to process, integrate, and visualize county-level origin–destination freight demand and revenue using confidential STB Waybill (*EBCDIC*) and FAF5 datasets, characterizing origin, interchange, and destination flows across Texas.
- Constructed internal and external rail network graphs and simulated regional rail corridors to quantify node and link performance functions for processing and travel times, deriving percentile-based performance distributions to represent operational variability.
- Built a statewide multimodal freight assignment framework integrating road, rail, and waterway networks, with generalized costs combining terminal processing, link travel time, operational cost, and risk.
- Developed multimodal hazmat exposure models and evaluated freight routing under hurricane, flooding, and link-closure disruptions to support resilience analysis and infrastructure investment planning.

National Laboratory of the Rockies

Research Assistant, Intermodal Freight Optimization (INFORMES, ARPA-E)

September 2023 – July 2026

Golden, CO

- Developed a SimPy-based discrete-event simulation framework (*LIFTS*) to evaluate intermodal terminal operations for *Parallel Systems* autonomous rail vehicles, modeling container flows, crane/hostler/vehicle interactions, congestion, and resource-constrained scheduling dynamics.

- Integrated mainline rail simulation outputs with terminal operations to quantify container processing times, terminal delays, resource utilization, and system-level operational performance; related work published in [Transportation Research Record](#).
- Developed geospatial data pipelines to extract and transform rail corridor attributes including grade, curvature, elevation, speed limits, and siding locations into standardized network inputs for [ALTRIOS](#), an open-source Rust- and Python-based rail simulation and optimization platform.
- Automated large-scale corridor simulations and analyzed system-level performance including train delay distributions, energy consumption, terminal processing times, and operational costs across corridor and nationwide scenarios.

National Rail Transit Testing Technology Laboratory

April 2020 – May 2022

Research Assistant, Track Crack Detection Model

Shanghai, China

- Developed image preprocessing, segmentation, and region-extraction pipelines using HALCON and Python for automated railway infrastructure inspection.
- Built an end-to-end computer vision system for railway track inspection using YOLOv3 and BASNet for object detection and surface-defect segmentation.
- Developed a PyQt-based visualization interface and contributed to the design of a camera-mounted autonomous railway inspection prototype.

Institute of Industrial Engineering, Tongji University

July 2021 – March 2023

Research Assistant, Machine Learning for Tumor Subregion Delineation and Prediction

Shanghai, China

- Developed an unsupervised clustering framework for clinical lung tumor imaging data, stratifying tumors into high-, intermediate-, and low-risk subregions and automatically delineating high-risk regions.
- Built time-series prediction models to forecast post-treatment changes in intermediate- and high-risk subregions, integrating segmentation and longitudinal prediction into clinical decision-support software for tumor delineation and treatment planning.

Shanghai Hongqiao International Airport

September 2020 – December 2021

Research Assistant, Collaborative Optimization Model for Airport Dispatching Decision

Shanghai, China

- Developed a stochastic dispatching model for airport ground transportation using M/M/S queuing theory to represent passenger–driver matching dynamics and service-capacity constraints.
- Built a VISSIM-based microscopic simulation framework to evaluate taxi routing strategies and terminal layout configurations under varying passenger-demand scenarios.
- Developed operational strategies balancing passenger waiting time, system throughput, service capacity, and driver income for airport ground transportation management.

Industry Experience

ByteDance

June 2023 – August 2023

Data Scientist Intern

Shanghai, China

- Built scalable video and image processing pipelines to extract and transform structured visual features for large-scale multimodal content understanding, model training, benchmarking, and evaluation.
- Integrated Segment Anything Model (SAM) into offline image and video preprocessing workflows, replacing object-detection-based visual features with region-level segmentation representations.
- Developed and evaluated Transformer-based multimodal vision models using embedding similarity and model-based evaluation methods, comparing representation quality and preprocessing cost for downstream short-video recommendation and search applications.

Ping An Technology

January 2023 – March 2023

Algorithm Engineer Intern

Shanghai, China

- Designed a multimodal image-captioning framework integrating image features, predicted semantic labels, and label-probability vectors to improve feature-importance modeling and image–text representation.
- Implemented a differentiable feature–label fusion mechanism enabling end-to-end backpropagation in generative language models and improving image–text alignment and caption accuracy over baseline training approaches.
- Extended the framework with Vision Transformer architectures to improve representation learning and generalization across application scenarios, contributing to a granted patent on image-description sentence generation systems.

Shanghai Shentong Metro Ltd

June 2021 – August 2021

Operation Management Intern

Shanghai, China

- Supported real-time metro operations at headquarters by coordinating with dispatchers, control center operators, and maintenance teams during daily operations and service disruptions.
- Monitored passenger-flow and operational data to identify congestion patterns and support scheduling, dispatching, incident response, and system performance tracking.
- Supported railway safety and maintenance activities by collecting and labeling track-surface crack data for the development of autonomous rail inspection equipment.

Technical Skills

Programming Languages: Python, Rust, C, C#, Julia, JavaScript, R, SQL, HTML

Optimization & Operations Research: Mixed Integer Linear Programming, Nonlinear Optimization, Convex Optimization, Large-Scale Network Optimization, Game Theory, Gurobi, CPLEX

Simulation & Transportation Modeling: Discrete-Event Simulation (SimPy), Microscopic Traffic Simulation (VISSIM), Agent-Based and Vehicle-Level Simulation (AnyLogic, OpenAI Gym), Track Design (AutoCAD, MicroStation), Queuing Models, Graph Theory, Network Routing/Assignment, Energy Modeling, Calibration and Sensitivity Analysis

Geospatial & Data Engineering: ArcGIS, GeoPandas, GeoPackages, Shapely, Fiona, Rasterio, DEM Processing, Graph Data Structures, Network Geometry, ETL Pipelines, Signal Alignment, Snowflake, Data Validation and Quality Assurance

Machine Learning & Artificial Intelligence: PyTorch, JAX, Scikit-learn, Reinforcement Learning, Physics-Informed Learning, Multimodal Representation Learning, Transformers, Computer Vision, Attention Mechanisms, Object Detection, Image Segmentation, LLM, RAG, Harness Optimization, Computational Resource Allocation

Visualization & Software: Tableau, JavaScript, PyQt, Dash, Git/GitHub, Cloudflare, MicroStation, AutoCAD

Publications

Peer-Reviewed Journal Articles

- [J8] **Tong, Q.**, et al. “Modeling the Competitiveness of Self-Propelled Autonomous Railcars in Short-Haul Intermodal Rail Service with User Equilibrium Network Assignment.” *Transportation Research Part A: Policy and Practice*. **Submitted**.
- [J7] **Tong, Q.**, et al. “Energy and Shipping Time Reduction Potential of Dynamically Dispatched Autonomous Electric Rail Vehicles” *Transportation Research Interdisciplinary Perspectives*. **Under-review**.
- [J6] **Tong, Q.**, et al. “Performance-based Mainline Travel Time Distribution Model for Railway Corridors in Support of Freight Network Analysis” *Journal of Rail Transport Planning & Management*. **Under-review**.
- [J5] **Tong, Q.**, Anderson, G., and Dick, C. T. “Sensitivity Analysis of Simulated Railway Energy Consumption to Elevation Data Processing and Train Length.” *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*. doi:10.1177/09544097261492758.
- [J4] **Tong, Q.**, and Dick, C. T. (2026). “Container Terminal Resource and Performance Comparison of Self-Propelled Autonomous Railcars and Conventional Intermodal Trains.” *Transportation Research Record*. doi:10.1177/03611981261476027.
- [J3] Wang, J., Li, L., Zheng, S., Zhao, S., Chai, X., Peng, L., Qi, W., and **Tong, Q.** (2023). “A Detection Method of Bolts on Axlebox Cover Based on Cascade Deep Convolutional Neural Network.” *Computer Modeling in Engineering & Sciences*, 134(3), 1671–1706. doi:10.32604/cmescs.2022.022143.
- [J2] Li, L., Zheng, S., Wang, C., Zhao, S., Chai, X., Peng, L., **Tong, Q.**, and Wang, J. (2022). “Crack Detection Method of Sleeper Based on Cascade Convolutional Neural Network.” *Journal of Advanced Transportation*, 2022, Article 7851562, 14 pp. doi:10.1155/2022/7851562.
- [J1] Zheng, D., Li, L., Zheng, S., Chai, X., Zhao, S., **Tong, Q.**, Wang, J., and Guo, L. (2021). “A Defect Detection Method for Rail Surface and Fasteners Based on Deep Convolutional Neural Network.” *Computational Intelligence and Neuroscience*, 2021, Article 2565500, 15 pp. doi:10.1155/2021/2565500.

Conference Papers & Presentations

- [C12] **Tong, Q.**, and Shang, Z. (2027). “Joint Model–Harness Optimization: Cost-Aware Resource Allocation for LLM Agents” *The 15th International Conference on Learning Representations (ICLR)*, San Francisco, USA, 2027.
- [C11] **Tong, Q.**, and Dick, C. T. (2027). “Performance-based Mainline Travel Time Distribution Model for Railway Corridors in Support of Freight Network Analysis.” *Transportation Research Board (TRB) 106th Annual Meeting*, Washington, D.C., USA, 2027.
- [C10] **Tong, Q.**, and Dick, C. T. (2027). “Energy and Shipping Time Reduction Potential of Dynamically Dispatched Autonomous Electric Rail Vehicles.” *Transportation Research Board (TRB) 106th Annual Meeting*, Washington, D.C., USA, 2027.
- [C9] **Tong, Q.**, and Dick, C. T. (2026). “Multi-Class User Equilibrium Model for Intermodal Freight Network Assignment.” *INFORMS Annual Meeting 2026*, San Francisco, USA, 2026. (**Invited Session**)
- [C8] **Tong, Q.**, Anderson, G., and Dick, C. T. (2026). “Sensitivity of Simulated Energy Consumption to Railway Corridor Elevation Data Processing and Train Length.” *INNOVA Rail 2026*, Champaign, IL, USA, 2026.

- [C7] **Tong, Q.**, and Dick, C. T. (2026). “Container Terminal Resource and Performance Comparison of Self-Propelled Autonomous Railcars and Conventional Intermodal Trains.” *Center for Transportation Research (CTR) Symposium*, Austin, TX, USA, 2026.
- [C6] **Tong, Q.**, and Dick, C. T. (2026). “Container Terminal Resource and Performance Comparison of Self-Propelled Autonomous Railcars and Conventional Intermodal Trains.” *Transportation Research Board (TRB) 105th Annual Meeting*, Washington, D.C., USA, 2026.
- [C5] **Tong, Q.**, and Dick, C. T. (2025). “Dynamic Routing for Intermodal Container Transportation.” *INFORMS Annual Meeting 2025*, Atlanta, GA, USA, 2025. (**Invited Session**)
- [C4] **Tong, Q.**, and Dick, C. T. (2025). “Optimal Layout Planning Method for Rail–Road Intermodal Container Terminals.” *11th International Conference on Railway Operations Modelling and Analysis (RailDresden 2025)*, Dresden, Germany, 2025.
- [C3] **Tong, Q.**, and Dick, C. T. (2024). “Intermodal Terminal Layout Optimization Considering Efficiency and Sustainability.” *INFORMS Annual Meeting*, Seattle, WA, USA, 2024.
- [C2] **Tong, Q.**, and Duan, C. (2023). “Voxelwise Two-Stage-Clustering and Machine-Learning Model for Robust Lung Cancer Tumor High-Risk Subregion Segmentation on FDG PET.” *INFORMS Annual Meeting*, Phoenix, Arizona, USA, 2023.
- [C1] Duan, C., Chen, S., Wang, J., **Tong, Q.**, Liu, Q., Yaseen, F., Wang, S., Hippe, D. S., and Bowen, S. R. (2023). “Two-Stage Clustering and Auto Machine Learning to Predict Chemoradiation Response in Tumor Subregions on FDG PET for LA-NSCLC.” *AAPM: The American Association of Physicists in Medicine*, Houston, Texas, 2023.

Patents & Software Copyrights

- [P3] “Multi-Target Detection Software for Motorways.” *National Copyright Administration*, 2022.
- [P2] “A ZigBee-Based Beacon System for Simulated Rail Traffic.” *National Intellectual Property Administration*, 2022.
- [P1] “Image Acquisition Device for the Inner Wall of Rail Transit Tunnels.” *National Intellectual Property Administration*, 2021.

Peer Review and Professional Service

Journal Referee

- Nature Communications, 1 review
- Transportmetrica B: Transport Dynamics, 1 review
- Transportation Research Record, 2 reviews
- Journal of Rail Transport Planning & Management, 10 reviews

Conference Referee

- Transportation Research Board Annual Meeting, 2 reviews
- IEEE INNOVA Rail 2026, 2 reviews

Professional Service

- Committee Member, Transportation Research Board Standing Committee (AT014) on Intermodal Freight and Truck Transportation, 2025–2027

Society Service

- President & Co-founder, **AREMA** Student Chapter at UT Austin, 2026–2027
- UT Student Transportation Council, 2026–2027
- Chinese Calligraphy Association, President, 2021–2022

Honors, Awards, and Scholarships

The University of Texas at Austin Fellowship	2023–2027
Outstanding Representative of National Scholarship Recipients , <i>Top 0.01% among all recipients</i>	2022
National Scholarship , <i>Ranked 1/407, GPA 4.0/4.0</i>	2021, 2022
Mathematical Contest in Modeling (MCM) , <i>Honorable Mention</i>	2022
APEC Innovation in Public Transport Competition , <i>Third Prize</i>	2021
Shenzhen Cup Mathematical Modeling Challenge , <i>Third Prize, Top 1% nationwide</i>	2021
Shanghai Innovation Award , <i>Excellent Innovation Report, Highest Award</i>	2021
National Mathematical Contest in Modeling , <i>Second Prize</i>	2020
Shanghai Scholarship	2020